

MOHAMMED USMAN F

AI / ML Engineer | GenAI, RAG & Deep Learning | Python • Bangalore, India

mohammedusmanf483@gmail.com | +91 93802 01698 | [linkedin.com/in/Mohammed-Usman-F](https://www.linkedin.com/in/Mohammed-Usman-F) | github.com/Usman1032

SUMMARY

Entry-level AI/ML Engineer (MCA, CGPA 8.7/10; AI-ML Engineer certified) who builds AI systems end to end, from data ingestion and embeddings to model integration, evaluation and API delivery. Built a RAG pipeline (FastAPI, ChromaDB, sentence-transformers, Claude API) with grounded, traceable outputs and graceful LLM fallbacks; a GPU-accelerated real-time inference service (TensorRT) with a 3.05x speedup over PyTorch at sub-4ms latency; and a Random Forest prediction platform served through Django. Solid foundation in deep learning (CNNs, attention, transformers) and Python backend engineering.

TECHNICAL SKILLS

GenAI & LLMs: Retrieval-Augmented Generation (RAG), Claude API integration, prompt engineering, grounded generation, agentic workflows, fallback and error handling for LLM calls

ML & Deep Learning: CNNs, attention, transformer architectures, loss functions, optimization, evaluation metrics; scikit-learn, Random Forest, feature engineering, model evaluation

Data & Vector Infra: ChromaDB, sentence-transformers embeddings, chunking & retrieval, data ingestion / preprocessing pipelines, Pandas, NumPy, SQL, SQLite, MySQL

Inference & Serving: FastAPI, Django, REST APIs, asyncio, server-sent events, TensorRT, CuPy, GPU-accelerated inference, Joblib model persistence

Programming & Tools: Python, JavaScript, Next.js / React, SQLAlchemy, Git, GitHub, Docker, Jenkins, CI/CD, Linux (working knowledge)

PROJECTS

AI Candidate Screening System – RAG-Powered Interview Simulator

2026

FastAPI, ChromaDB, sentence-transformers, Claude API, Next.js, SQLAlchemy | [GitHub](#) | [Demo video](#)

- Built an end-to-end RAG pipeline: resume parsing and profile extraction, chunking with overlap, local sentence-transformer embeddings, and retrieval from per-role ChromaDB collections.
- Generated interview questions and summaries grounded in retrieved context using the Claude API, with deterministic fallbacks so the pipeline keeps working if the LLM call fails.
- Designed traceability by storing each question's retrieval query and source chunks, and adapted question difficulty from answer-quality signals.
- Exposed the system through a layered FastAPI REST backend (routers / services / RAG) with a Next.js front end; documented architecture and trade-offs.

VisionEdge – Hardware-Accelerated Real-Time Video Pipeline

2026

Axlero Solutions Intern Project Program | Python, FastAPI, TensorRT, CuPy, PyAV, aiortc, React

- Designed and tested a FastAPI backend for a GPU-resident real-time object detection pipeline, combining hardware-accelerated video decode with TensorRT-optimized inference.
- Achieved a 3.05x speedup over native PyTorch with sub-4ms latency using async orchestration for concurrent multi-stream inference.
- Built a server-sent events API for live telemetry to monitor throughput and reliability.

AI-Powered Real Estate Investment Decision Platform

2025 – 2026

Python, scikit-learn, Pandas, NumPy, Django, SQLite/MySQL, Joblib | [GitHub](#)

- Trained Random Forest models for property value and rental income prediction (feature engineering on area, BHK, age, floor ratio, location), persisted with Joblib and served through a Django backend.
- Built a finance engine (EMI, ROI, affordability, investment score) and a BUY / HOLD / AVOID analysis report with charts, backed by user, property and analysis database models.

EDUCATION

Master of Computer Applications (MCA) – Sri Siddhartha Institute of Technology, Tumkuru

Sep 2024 – Aug 2026

CGPA: 8.7 / 10

BSc in Computer Science – Government First Grade College, Tumkuru

Oct 2021 – Sep 2024

CGPA: 7.66 / 10

CERTIFICATIONS

- AI-Machine Learning Engineer Certificate (150 hrs) – Reliance Foundation Skilling Academy, Skill India Digital (NSDC) – Aug 2026
- Django Full Stack Development – SSIT, Tumkuru – May 2025
- Junior Software Developer (SSC/Q0508), NSQF Level 3 – Skill India / NASSCOM IT-ITeS SSC – Jul 2024